

Southeast University

Department of Computer Science & Engineering

Course Code & Title: CSE 472 - Web and Internet Programming

Assignment Title: Individual Project - Full-Stack Mini Web Application

Weightage: 40% of total marks (40 marks)

Submission Type: Individual Assignment

Submission Deadline: TBA

1. Introduction

As part of this course, you are required to complete a comprehensive individual project that demonstrates your ability to design and implement a complete web application integrating frontend, backend, and database technologies. This assignment is worth 15% of your final grade and is designed to evaluate both your technical skills and your independent problem-solving abilities.

2. Objectives

The objectives of this assignment are to:

1. Enable you to apply knowledge of HTML, CSS, JavaScript, backend programming (e.g., PHP), and databases (e.g., MySQL) to develop a functional web application.
2. Provide experience in designing and implementing user authentication and CRUD operations in a secure, efficient manner.
3. Encourage you to analyse problems and translate requirements into a working system independently.
4. Foster project management and documentation skills by planning the development process, documenting your work, and presenting the final results.

3. Project Requirements

You will construct a full-stack web application on a theme of your choice, such as an Event Registration System, Online Library, Student Portal, Product Catalogue, or Task Tracker.

Your application must include:

- **Pages:** Minimum 4-5 interconnected pages (Home, Login/Registration, CRUD Page, and one additional page of your choice).
- **Frontend:** HTML, CSS, and JavaScript for structure, layout, and interactivity.
- **Backend:** Recommended PHP, but any equivalent backend technology may be used.
- **Database:** Recommended MySQL, but any relational/NoSQL database is acceptable.
- **Authentication:** User registration, login, logout with secure password hashing.
- **CRUD Operations:** Create, Read, Update, Delete for one main entity (e.g., product, book, task, event).
- **Search/Filter:** At least one simple search or filter function.
- **Validation & Security:** Input validation, prevention of SQL injection/attacks, and basic access control (sessions/tokens).

4. Technology Flexibility & Hosting

- The recommended stack is HTML, CSS, JavaScript, PHP, and MySQL, as taught in this course. However, you may use any equivalent stack (e.g., Node.js + MongoDB, Django + PostgreSQL, Laravel, etc.) if you prefer. All functional requirements must still be met.

- **Code Repository:** The project source code must be uploaded to a personal GitHub repository. A clear `README.md` file with setup instructions is mandatory.
- **Live Hosting:** The website must be hosted live. You can use GitHub Pages or Netlify (for static/frontend sites), or a suitable free hosting platform (e.g., Render, InfinityFree, Vercel) depending on your backend technology.

5. Deliverables

You are required to make one single submission on the LMS:

Documentation Report (.pdf) To be submitted on the LMS under the "Assignment - Documentation" link.

- **File name:** `StudentID_CSE471_Assignment.pdf`

The report (approx. 8-10 pages) must include:

- **Cover Page:** Course, title, your name & ID. Must also include the GitHub Repository Link and the Live Website Link.
- **Objectives & Scope:** A brief overview of the application.
- **Architecture Diagram:** Client → Server → Database flow.
- **Database Schema:** ER diagram, table definitions, keys.
- **Features:** Screenshots of login, CRUD, search, etc., with explanations.
- **Validation & Security:** Notes on how secure coding was ensured.
- **Testing:** Checklist of at least 5 test cases (valid, invalid, edge cases).
- **Development Timeline:** A brief log or table showing how you managed your time across frontend, backend, database, and documentation phases.
- **Reflection:** Short write-up on lessons learned and challenges faced during independent development.
- **References:** Properly cited in Harvard style.

6. Independent Work Policy

- This is strictly an individual assignment.
- You are entirely responsible for all aspects of the project (frontend, backend, database, and documentation).
- Sharing code with classmates or collaborating on the actual development is strictly prohibited and will be treated as an academic integrity violation.

7. Academic Integrity

- Plagiarism from other students' projects or copying from online sources without citation will result in zero marks.
- Proper citation must be provided for any borrowed content (images, icons, boilerplate code snippets).

- **Plagiarism Policy:** A similarity index above 20% will result in mark deductions depending on severity. In serious cases (copy-paste or student-to-student copying), the assignment may be awarded zero (0) marks. All work must be written in your own words using proper Harvard-style referencing.

8. Submission Instructions

- You must submit ONE file (PDF) on Google classroom.
- **Filename:** `StudentID_CSE471_Assignment.pdf`
- The PDF must include all required sections, and most importantly, the working links to your GitHub Repository and your Live Hosted Website on the cover page.
- Do not upload ZIP files of your code to the LMS; your code will be evaluated directly from your GitHub repository.
- **Deadline:** TBA. Late submissions follow department policy.
- **Presentation (mandatory):** You must be present on the scheduled day to demonstrate your project and answer questions on any part of your work. Non-attendance may result in a mark penalty or zero for the assignment.